

Jack Lockhart

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EDUCATION

The University of Alabama

Master of Business Administration (MBA)

Tuscaloosa, AL

Expected May 2028

- GMAT Focus Edition: 715 (99th Percentile)

Bachelor of Science in Cyber Security | GPA: 4.00/4.00

Expected May 2027

Bachelor of Science in Computer Science | GPA: 4.00/4.00

Expected May 2027

EXPERIENCE

Undergraduate Research Assistant, Agentic AI Trust

May 2026 – July 2026

University of Alabama

- Developed a neural-network-based framework to model and estimate human trust in agentic AI systems by fusing structural, temporal, and behavioral interaction data.
- Engineered multi-stream data pipelines to process behavioral metadata, including keystroke dynamics, pause times, and sentiment signals via fine-tuned BERT models.

Electrical Team Member / Software Engineer

Sep 2023 – Present

Baja Racing

- Programmed an Arduino-based system to collect and manage data from over \$2,000 of onboard sensors.
- Collaborated with cross-functional electrical and mechanical teams to design, prototype, and implement hardware and software solutions for vehicle performance optimization.

PROJECTS

Building Energy Consumption Predictor | *Python, Machine Learning, Data Analytics, API integration*

Campus Energy Consumption Forecasting Tool

- Engineered an end-to-end data pipeline processing 28M+ energy records across 1,000+ buildings, using modified Z-score outlier filtering, gap interpolation, and real-time weather API integration.
- Designed domain-specific thermal features (3-hour rolling temperature, Cooling Degree Hours) and tuned an XGBoost regressor via Optuna (Bayesian Optimization) to capture non-linear HVAC activation thresholds.
- Reduced prediction error from 64.9% to 18.5% NMAE compared to baseline linear models and deployed the trained model as an interactive web dashboard.

Password Manager & Generator | *Cryptography, Password Security*

Local Encrypted Credential Vault

- Built a local encrypted credential vault using AES-256-GCM with PBKDF2-SHA-256 key derivation, ensuring vault keys are never written to unencrypted storage.
- Integrated dual real-time password strength meters showing Bitwarden's zxcvbn score compared to custom heuristics (entropy, length, sequential analysis, and dictionary matches).

CERTIFICATIONS

- CompTIA Security+
- CompTIA Network+

CAMPUS ACTIVITIES

- Crimson Defense Cyber Security Team (2023 – present)
- Baja Racing club (2023-present)
- Club Tennis Team

SKILLS

Security & Networking: Wireshark, Nmap, Cryptography, Linux, Networks, Virtual Machines

Cloud & DevOps: AWS (IAM, Lambda, S3, CloudFront), Microsoft Azure, Oracle Cloud, Docker, Git, CI/CD

Languages: Python, Java, C++, JavaScript, TypeScript, Bash